UNIT 7 · RAPID REVISION

Formula and Fact Sheet

Solar energy

Item	Value
Solar constant	1367 W/m² (WMO) ≈ 4.92 MJ/m²·h; modern value ~1361 W/m²
Annual variation of the solar constant	±3.3% (maximum early January)
Sun as a black body	~5800 K; Wien peak ~0.48 μm
Spectrum shares	UV ~7%, visible ~47% (PAR 0.4–0.7 μm), IR ~46%
Air mass	AM = 1/cos θz; AM0 outside, AM1.5 PV standard (~1000 W/m²)
Pyranometer / pyrheliometer	Global radiation / beam radiation (tracks the sun)
Diffuse radiation	Pyranometer + shade ring
Campbell–Stokes recorder	Hours of bright sunshine
India insolation	~5–7 kWh/m²/day, ~300 sunny days
Flat-plate collector	Global radiation, no tracking, ≤ ~100 °C
Concentration ratio	CR = aperture area ÷ receiver area
Silicon band gap / cell voltage	1.1 eV / 0.5–0.6 V
Fill factor	FF = (V_mp·I_mp)/(V_oc·I_sc)
STC	1000 W/m², 25 °C, AM1.5
PV temperature coefficient	~0.3–0.5% output loss per °C

Wind and biomass

Item	Value
Wind power	P = ½ρAV³ (×Cp extractable)
Betz limit	Cp(max) = 16/27 = 0.593
Real turbine Cp	0.35–0.45
Cut-in / rated / cut-out	~3–4 / ~11–14 / ~25 m/s
Producer gas	CO + H_2 + CH_4; CV ~4–6 MJ/m³
Fast vs slow pyrolysis	Fast → bio-oil; slow (carbonization) → charcoal
Loose residue vs densified	40–150 vs 600–1200 kg/m³
Briquette / pellet size	~60–90 mm / ~6–12 mm
Densification moisture	~8–12%

Bioconversion and energy analysis

Item	Value
Raw biogas	55–65% CH_4, 35–44% CO_2; CV ~20–26 MJ/m³
Digestion optimum	~35 °C, pH 6.8–7.2, C:N 20–30:1
CBG (IS 16087:2016)	≥ 90% methane, ~250 bar
KVIC vs Deenbandhu	Floating drum (constant pressure) vs fixed dome (no moving part, variable pressure)
Biodiesel	Transesterification: oil + methanol (KOH) → methyl esters + glycerol
Energy ratio	Output ÷ input
Specific energy	Input ÷ yield (MJ/kg)
Energy equivalents	Diesel ~56 MJ/L; labour ~1.9 MJ/h; bullock pair ~10 MJ/h; electricity ~11.9 MJ/kWh

Energy sources and solar radiation (Topics 1–2)

- Conventional = finite fossil/nuclear; renewable = solar/wind/biomass/hydro/tidal/geothermal + muscle. Wind, biomass and hydro are INDIRECT solar. Fertiliser = largest indirect farm energy.
- Solar constant = 1367 W/m² (WMO), ≈ 1.94 cal/cm²·min; ±3.3% yearly (maximum early January); sun ≈ 5800 K black body; Wien peak ~0.48 μm.
- Spectrum: UV ~7%, visible ~47% (PAR 0.4–0.7 μm), IR ~46%; AM = 1/cos θz; AM0 space, AM1.5 standard (~1000 W/m²).
- Global = beam + diffuse (+ reflected on tilt). Pyranometer = global; pyrheliometer = beam (tracks the sun); pyranometer + shade ring = diffuse; Campbell–Stokes = sunshine hours.
- India: RE 288.58 GW, solar 162.15 GW (30 June 2026); 500 GW non-fossil by 2030; PM-KUSUM A plants / B standalone pumps / C solarisation.

Solar collection and PV (Topics 3–4)

- FPC: glazing (greenhouse effect) + black absorber + insulation; global radiation, no tracking, ≤ 100 °C; $\eta = F_R(\tau\alpha) - F_R U_L(T_i - T_a)/I$; $Q_u = \dot{m}c_p\Delta T$.
- Concentrators use BEAM only + TRACK; CR = aperture/receiver; trough/Fresnel (line, 200–400 °C), dish/tower (point, > 500 °C).
- Storage: sensible ($Q = mc_p\Delta T$; water, rock bed), latent (PCM, constant T), thermochemical.
- PV: p–n junction; Si band gap 1.1 eV, cell ~0.5–0.6 V; series adds V, parallel adds I; MPP, FF; STC 1000 W/m², 25 °C, AM1.5; output ↓ ~0.4%/°C.
- Mono > poly > thin-film; 2026: PERC 21–22%, TOPCon 22–23%, HJT ~22–27%, tandem lab ~34.8%. Battery-less solar pump (store water, not charge) is cheapest.

Wind and thermo-chemical biomass (Topics 5–7)

- $P = \frac{1}{2}\rho AV^3$; extractable = $\frac{1}{2}\rho AV^3 \cdot Cp$; BETZ $Cp(max) = 16/27 = 0.593$; $P \propto V^3$ (2× speed → 8× power) and $\propto D^2$; λ = tip speed/wind speed.
- HAWT (propeller, multi-blade pump) needs yaw; VAWT (Darrieus lift, Savonius drag) omni-directional; cut-in ~3–4, rated ~12, cut-out ~25 m/s; India wind ≈ 56 GW (4th).
- By oxygen: combustion (excess air → heat), gasification (partial → producer gas CO + H_2 + CH_4, 4–6 MJ/m³), pyrolysis (none → char + bio-oil + gas), carbonization (slow pyrolysis → charcoal).

- Densification: briquette 60–90 mm, pellet 6–12 mm; 600–1200 kg/m³; lignin binds under heat/pressure; moisture ~8–12%; co-firing cuts stubble burning.

Bioconversion, alternate fuels and energy analysis (Topics 8–12)

- Anaerobic digestion: hydrolysis → acidogenesis → acetogenesis → methanogenesis; ~35 °C, pH 6.8–7.2, C:N 20–30; biogas 55–65% CH_4 + 35–44% CO_2, CV ~20–26 MJ/m³.
- KVIC floating drum (constant pressure, steel corrodes) vs Deenbandhu fixed dome (no moving part, cheap, variable pressure).
- Ethanol by fermentation (E20); biodiesel by TRANSESTERIFICATION (triglyceride + methanol/KOH → methyl esters + glycerol); CBG (IS 16087:2016) ≥ 90% CH_4 at 250 bar; SATAT.
- Alt fuels: ethanol (octane ↑, CV ↓, SI), biodiesel (cetane ↑, CI), biogas/CNG, producer gas, H_2, electric; issues = material compatibility, CV drop, storage.
- Direct vs indirect energy (fertiliser largest); energy ratio = output/input; specific energy = input/yield (MJ/kg); energy audit = inventory of all in/out flows in MJ.

Further reading

- Sukhatme & Nayak — Solar Energy: Principles of Thermal Collection and Storage.
- Duffie & Beckman — Solar Engineering of Thermal Processes; Rai — Non-Conventional Energy Sources.
- Mittal — Energy in Agriculture (energy equivalents, energy ratio, energy audit).
- MNRE physical progress data and scheme guidelines (PM-KUSUM, PM Surya Ghar, National Bioenergy Programme, SATAT).
- IS 16087:2016 (Compressed Bio-Gas specification); WMO Guide to Meteorological Instruments (radiation).
